WHITE PAPER · HWACHEON HI-ECO 35 (TWO CELLS) × INDUSTRIAL & GENERAL MANUFACTURING

Hwacheon Hi-Eco 35 for Industrial & General Manufacturing

Mid-size CNC lathe pair for shafts, gear blanks, flanges, and threaded components with consistency and clean finishes. This paper covers what the machine does, the industrial & general manufacturing parts we produce with it, the alloys we run, the tolerances and finishes industrial & general manufacturing buyers expect, and the documentation package.

12.5" OD × 42" L max turning (each cell) ENVELOPE	**±0.0005" routine on critical features** TOLERANCE
1994 SINCE	**Same-week / rig-down direct** RESPONSE



Ironwood Revival / Quickdraw Slim · 12.5" OD × 42" L max turning (each cell) · mid-size precision CNC turning center

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Executive summary

Machine: Hwacheon Hi-Eco 35 (two cells) (shop nickname: *Ironwood Revival / Quickdraw Slim*). mid-size precision CNC turning center at B&R Productions, running from our New Waverly, Texas shop.

Application: Mid-size CNC lathe pair for shafts, gear blanks, flanges, and threaded components with consistency and clean finishes for Industrial & General Manufacturing customers — Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

Envelope: 12.5" OD × 42" L max turning (each cell). **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

The Hi-Eco 35 pair is the mid-size backbone of the shop's turning capacity. Two identical cells means when one is loaded with a long-cycle Inconel job, the other keeps repair and small-batch work moving. Shafts, gear blanks, flanges, threaded components — the class of parts that make up the bulk of oilfield turning volume runs here.

Why Ironwood Revival / Quickdraw Slim for Industrial work

Mid-size industrial turned parts — shafts, gear blanks, flanges, bushings, threaded components — are bread-and-butter production work, and two identical 12.5" × 42" cells are how you deliver them on a date. Redundancy protects the schedule; identical programs make the redundancy real.

What this looks like on a real part

A repeat run of flanged shafts is a consistency problem more than a difficulty one. The hundredth part has to match the first, which means tool wear, thermal drift, and workholding repeatability all have to be controlled across the run rather than measured at the end of it. Splitting a run across two cells only works if both cells are genuinely set up the same way — same tooling, same offsets — which is a housekeeping discipline, not a machine capability.

When this is the wrong machine

42 inches. Longer industrial shafting goes to Rattle & Hum; parts needing milled features in the same setup go to Torque & Tumble.

Full specifications

Model	Hwacheon Hi-Eco 35 — two cells: #16 ("Ironwood Revival") + #14 ("Quickdraw Slim")
Type	Mid-size CNC turning centers, run in parallel for throughput
Max turning capacity	12.5" OD × 42" L per cell (B&R working spec)
Platform max swing	23.6" over bed; 16.35" turning diameter platform-max
Platform max length	50" between centers
Chuck	12" 3-jaw hydraulic (15" available on some builds)
Turret	10-station
Control	Fanuc OT-C
Configuration	Two identical cells for parallel throughput
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, carbon steel, aluminum, Inconel, Duplex, 17-4 PH, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Industrial & General Manufacturing

Typical buyer: Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

What's hard about industrial & general manufacturing work: Legacy parts with no OEM support, one-off replacement components, small-lot production runs, materials that don't fit standard shop capabilities, and tight lead times on repair work where a down-machine is losing money by the day.

Typical Industrial & General Manufacturing parts we produce on Ironwood Revival / Quickdraw Slim

- Mid-size shafts, spindles, and precision round parts
- Pump components (impellers, sleeves, seal areas)
- Valve stems and cylinders
- Rollers, bushings, bearings, adapters
- Reverse-engineered legacy replacement parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Industrial & General Manufacturing

Standard range: Stainless (304, 316, 17-4 PH) · Tool steels · 4140 / 4340 · Brass · Bronze · Aluminum · Cast iron · Specialty alloys per customer requirement.

Industrial work covers the widest alloy range because the applications are the widest. Food and pharma often specify 316L or 17-4 PH; mining and heavy equipment specify tool steels and 4140/4340; pump manufacturers outside oilfield use everything from bronze bushings to stainless impellers. We work with the customer's spec sheet — if you tell us the alloy, we'll tell you honestly whether we can machine it and how quickly.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, industrial & general manufacturing or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Industrial & General Manufacturing

±0.0005" routine on critical features. Industrial applications range from precision (pump seal areas, close-fit bushings) to structural (mining and heavy-equipment brackets); we adjust the inspection plan to what the part actually needs. Surface finish measured on parts where finish drives function.

Documentation and quality workflow

Documentation calibrated to customer requirement. Material traceability standard for critical parts. Certificates of conformance issued with delivery. NDAs on request for proprietary designs or reverse-engineered legacy parts.

Response time and emergency work

Down-machine and production-critical work prioritized. Lead times honestly quoted — no false promises. For legacy reverse-engineering, we typically confirm feasibility on the first phone call. Call (936) 291-7827.

How we compare

Compared to sourcing a new OEM part: we're often faster and cheaper for one-offs and small runs, and for legacy parts where the OEM has moved on we're often the only realistic option. Compared to a cheapest-bidder job shop: we hold tolerance, document material, and don't ruin the part.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Hi-Eco 35 platform and Industrial & General Manufacturing work. Also indexed as FAQ schema for AI answer engines.

Why does B&R run two Hi-Eco 35 cells instead of one?

Parallel throughput. When one cell is loaded with a long-cycle Inconel job, the other keeps mid-size repair and small-batch work moving. Two identical cells means predictable capacity when volume matters.

What's the max turning capacity per cell?

12.5" OD × 42" L is B&R's working spec. The platform max is 16.35" turning diameter × 50" between centers with a 23.6" swing.

What chuck size and turret does the Hi-Eco 35 use?

12" 3-jaw hydraulic chuck (15" available on some builds). 10-station turret. Fanuc OT-C control.

What kind of industrial customers does B&R work with?

Industrial equipment OEMs, custom machinery builders, non-oilfield pump and valve manufacturers, food and pharma equipment suppliers, mining and heavy-equipment operators, and any customer who needs a legacy or reverse-engineered part made right.

Do you machine one-off replacement parts?

Yes — a lot of our industrial work is exactly this. Send the sample or drawing; if the material is on the shelf, we're often faster than the OEM. Reverse-engineered legacy parts across almost every sector.

What materials do you run for industrial work?

Full range — stainless (304, 316, 17-4 PH), tool steels, 4140/4340, brass, bronze, aluminum, cast iron, and specialty alloys per customer requirement. If you need something unusual, ask.

Do you offer emergency turnaround on down-machine parts?

Yes for repeat customers with material on the shelf. Same-day and next-day realistic when we already have the alloy. Call (936) 291-7827 and describe what's down.

Can you do small-batch production or is B&R just one-offs?

Both. Prototype and small-batch through repeat production. Small production runs where a big shop's minimum order would kill the economics are exactly where we fit.

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Ready to talk about your Industrial & General Manufacturing work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.

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